Data Science PORTFOLIO

Mochammad Fahd Ali Hillaby | 2026

M. FAHD ALI HILLABY

Data Science Technology Student

Hello, my name is Aby

I am currently pursuing a fresh graduate in Bachelor's degree in Data Science Technology. I have a strong interest in the fields of data analysis and statistics. I am passionate about leveraging the potential of data and advanced technology to address complex challenges, uncover valuable insights, and develop innovative solutions. Through my academic journey, I aim to deepen my understanding of data-driven approaches and contribute to impactful projects that make a difference in various industries.

EDUCATION

Bachelor of Data Science Technology
Airlangga University
2022 - now

ACTIVITIES

BEM FTMM 2023
Dirjen Arttaiton Medkominfo
2023

BEM FTMM 2024
Inspectorat General
2024

IRIS 2024
Researcher
2024

IRIS 2025
Head of Public Relation
2025

EXPERIENCE

PT Semen Indonesia
CSR Data Analyst Intern
2026

PT Indospring, tbk
Human Capital Data Analyst
2025

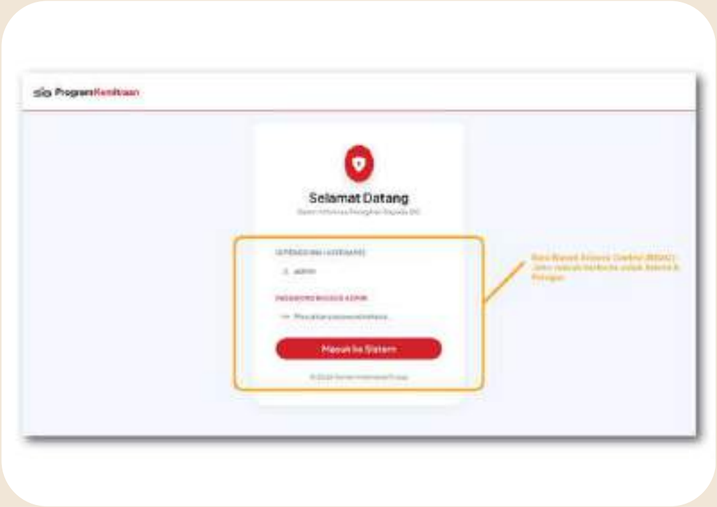
ICATAM 2024
Contribute Paper
2024

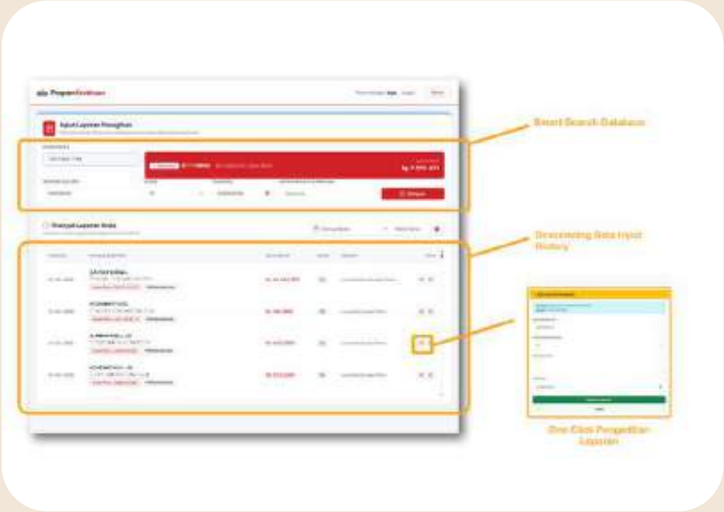
DATA ANALYSIS PROJECT'S

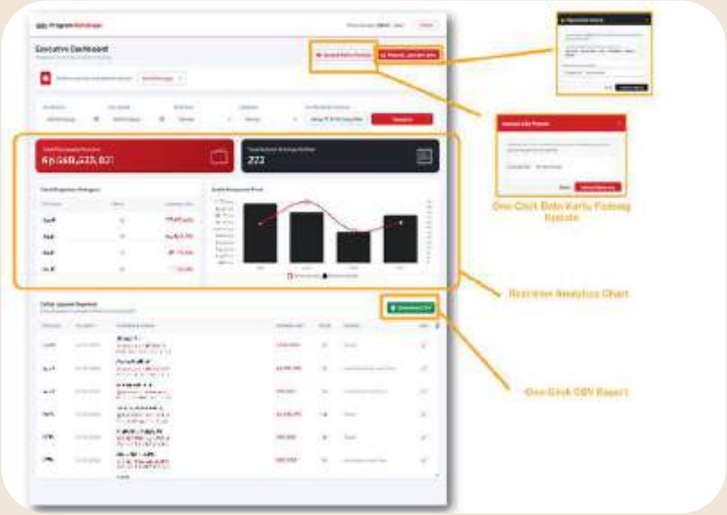
2023-2026

Mochammad Fahd Ali Hillay | 2026

Project Overview: Integrated CSR & COMDEV Reporting System







A full-stack web application designed to digitize and centralize the reporting process for CSR and Community Development programs at PT Semen Indonesia (Persero) Tbk (SIG). Built using Next.js and a serverless PostgreSQL database, this system replaces traditional, fragmented spreadsheet-based workflows.

Topic :

- Full-Stack Web Development
- Database Management & ORM
- Data Pipeline & Automation
- Executive Data Visualization

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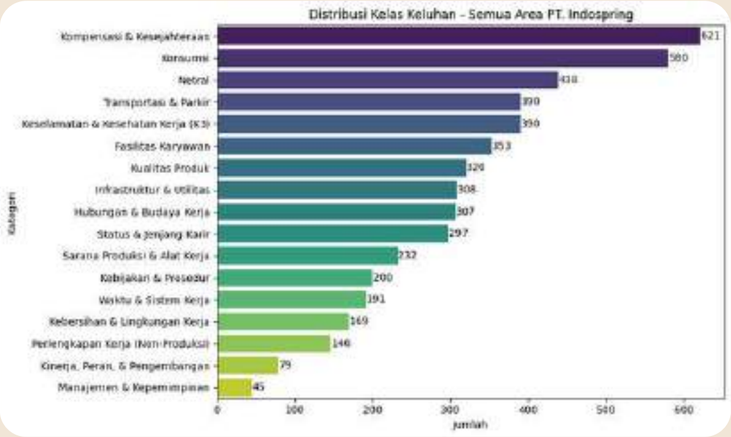
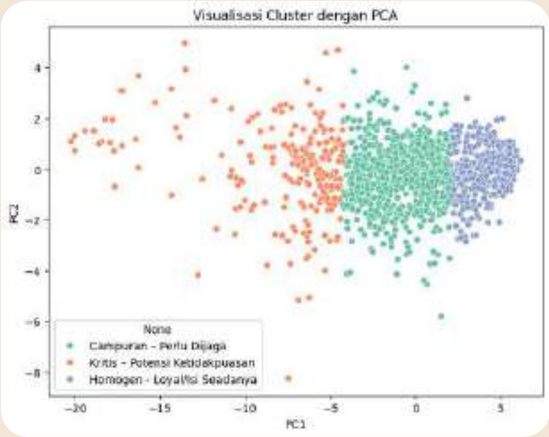
Sentiment Analysis and Employee Segmentation of Employee Opinion Survey at PT. Indospring Tbk. Using NLP and K-Means Clustering.





During my internship at PT. Indospring Tbk., I conducted a Data Science project focused on optimizing Human Capital management by analyzing employee feedback. This project integrated Natural Language Processing (NLP) and K-Means Clustering to transform raw survey responses into actionable insights for enhancing corporate engagement and satisfaction strategies.

Topic :

- Sentiment Analysis of Employee Opinions
- Employee Segmentation
- Strategic Recommendations

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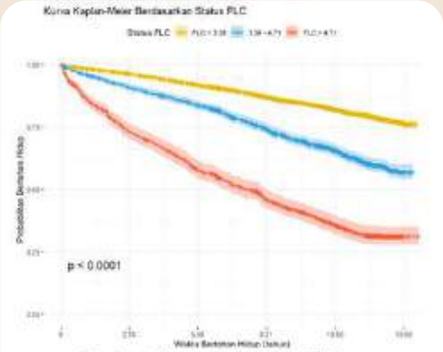
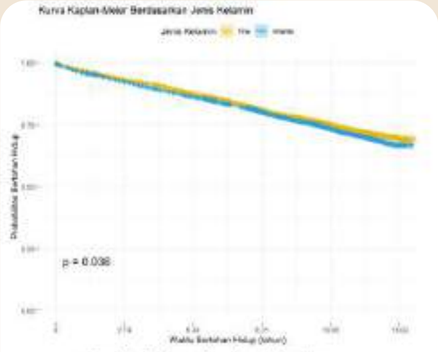
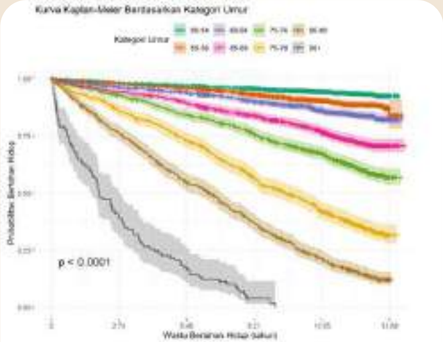
Analysis of Survival Based on Serum Free Light Chain and Other Risk Factors



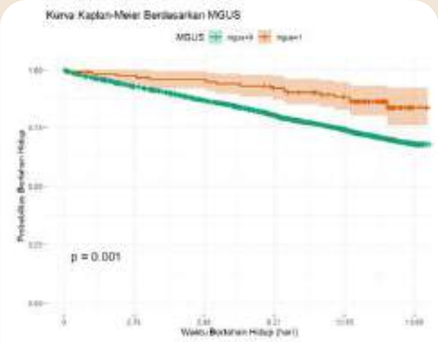
Gambar 1. Kurva Kaplan Meier FLC



Gambar 3. Kurva Kaplan Meier Sex



Gambar 2. Kurva Kaplan Meier Age



Gambar 3. Kurva Kaplan Meier MGUS

This project analyzed the relationship between serum free light chain (FLC) levels and risk factors in predicting mortality using the flchain dataset from a population study in Olmsted County, Minnesota. We explored FLC as a survival biomarker alongside variables like age, gender, and MGUS status.

The analysis involved data cleaning, exploratory analysis, and Cox Proportional Hazard (CoxPH) modeling. To handle proportional hazards (PH) assumption violations, we applied stratified and extended Cox models, with the stratified model with interaction achieving the best fit (AIC: 30,646.04).

Topic :

- Survival Analysis
- Cox Proportional Hazard Modeling
- Stratified and Extended Cox Models
- Biomarker Analysis
- Exploratory Data Analysis (EDA)

--> Link Projek <--

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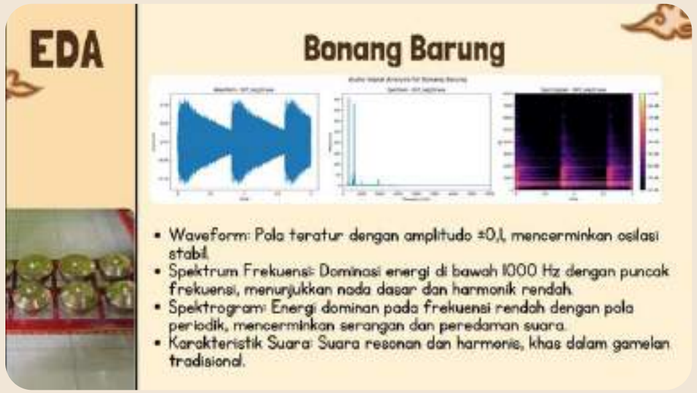
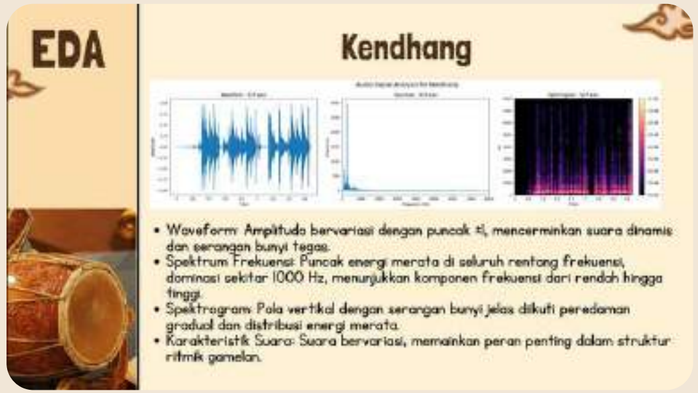
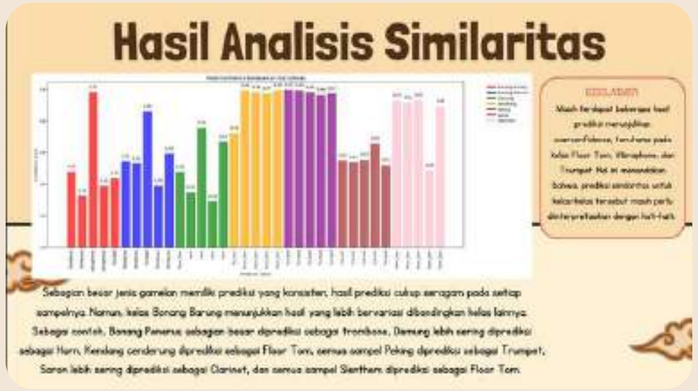
Gamelan Instrument Classification Using Audio Features and Machine Learning







This project focused on classifying gamelan instruments using audio features to support the preservation of Indonesian cultural heritage. We analyzed audio recordings, extracted features using MFCC, and tested various machine learning models. Random Forest and LSTM achieved the highest accuracy at 96%, with CNN and XGBoost also performing well. I contributed to feature analysis, model development, and evaluation in this collaborative project.

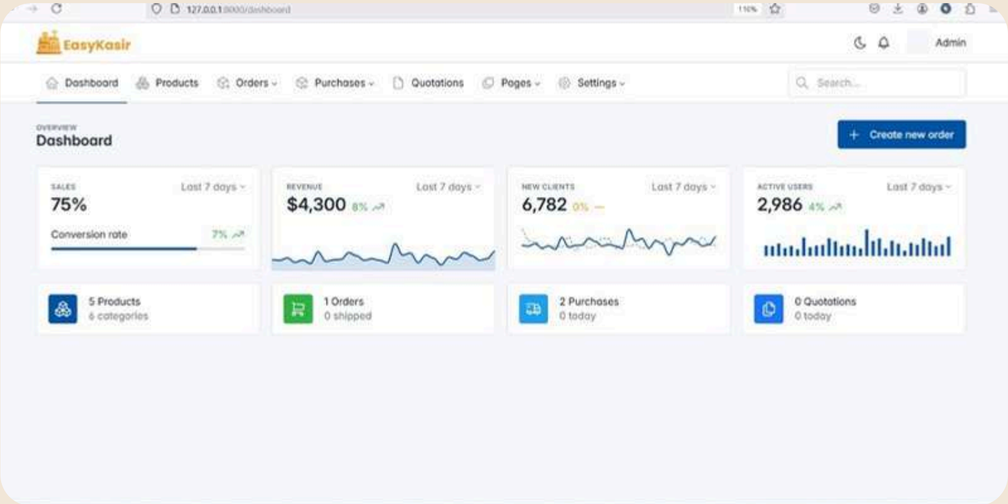
Topic :

- Audio Feature Extraction
- MFCC (Mel-Frequency Cepstral Coefficients)
- Machine Learning Classification
- Deep Learning (LSTM, CNN)
- Sound Embedding Analysis
- Exploratory Data Analysis (EDA)

--> Link Projek <--

Mochammad Fahd Ali Hillay | 2026

Implementation of Smart Cashier System Using Laravel



This project focused on developing EasyKasir, a web-based Smart Cashier System, to enhance transaction management and inventory control for small and medium-sized businesses. The system addresses the inefficiencies of traditional cashier methods, offering features like inventory tracking, sales reporting, and real-time notifications to improve operational productivity.

The system includes features for user management (admin, cashier, customer roles), inventory control, supplier management, and comprehensive sales reports. It also features a user-friendly dashboard with night mode and customizable themes for better usability.

--> Link Projek <--

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Queueing System Analysis at a Barbershop – A Case Study of The Magic Barbershop, Surabaya

This project analyzes the efficiency of the queueing system at The Magic Barbershop, Keputih, Surabaya, using a Multi-Channel Single Phase Queue model (M/M/s). Observations were conducted during weekdays and weekends, both in the afternoon and evening, to evaluate variables such as arrival and service rates, waiting time, idle probability, and system utilization.

The analysis results show that evenings, especially on weekends, have the highest arrival rates and system utilization, yet remain efficient in terms of customer waiting time. The resulting recommendations include adjusting the number of staff and optimizing service time to improve customer satisfaction.

This project used Python and Google Spreadsheet for modeling and statistical analysis, providing practical insights into applying queueing theory and stochastic processes in real-world service settings.

Skills: Stochastic Methods · Statistics · Data Analysis

--> Link Projek <--

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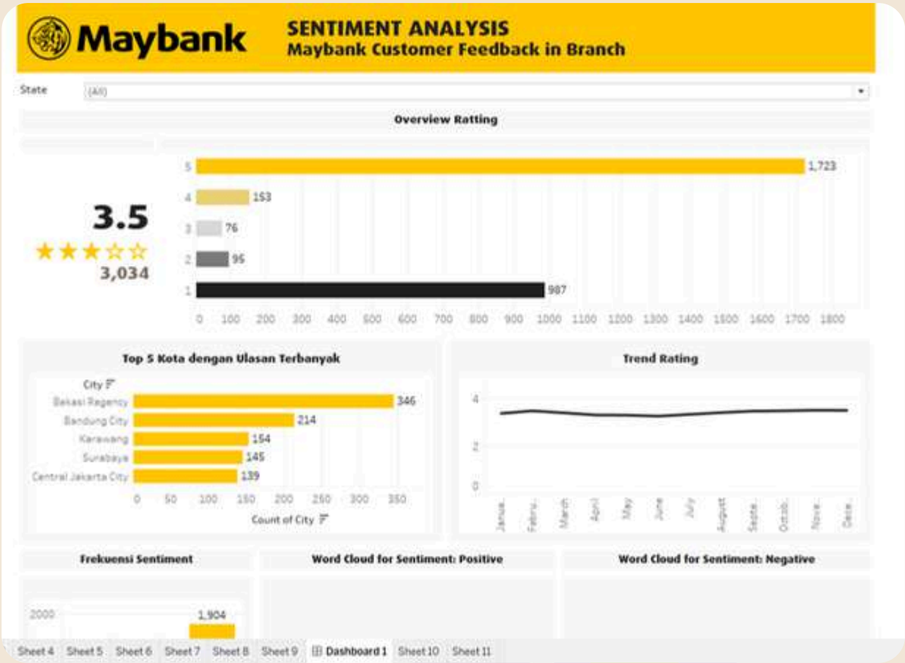
Sentiment Analysis Review Maybank Customer in Branch



This project was part of the Data Science Consulting program (SDC22), where I joined a data analyst team to solve a real-world case study. My role in the team was as a visualization specialist, responsible for designing and building an interactive dashboard using Tableau.

The dashboard aimed to present insights from [specify data type if relevant, e.g., sales, customer satisfaction, etc.], supporting data-driven decision-making.

In this work, I applied effective visual design principles, built interactive layouts, and leveraged Tableau features such as dynamic filters and comparative charts to make the data easier for end users to interpret.

Skills: Data Visualization · Tableau · Data Analysis

--> Link Projek <--

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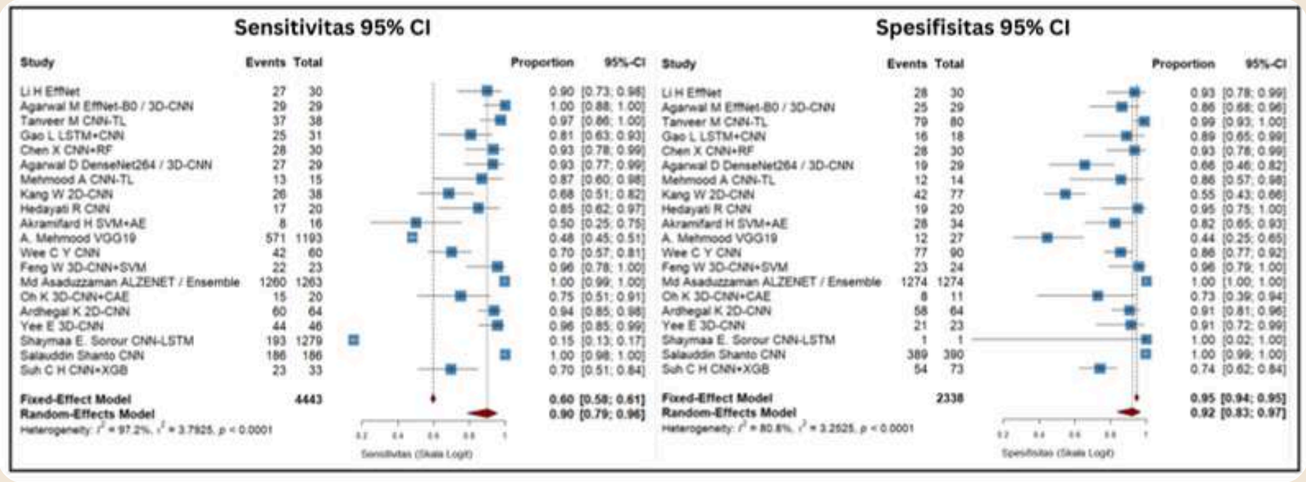
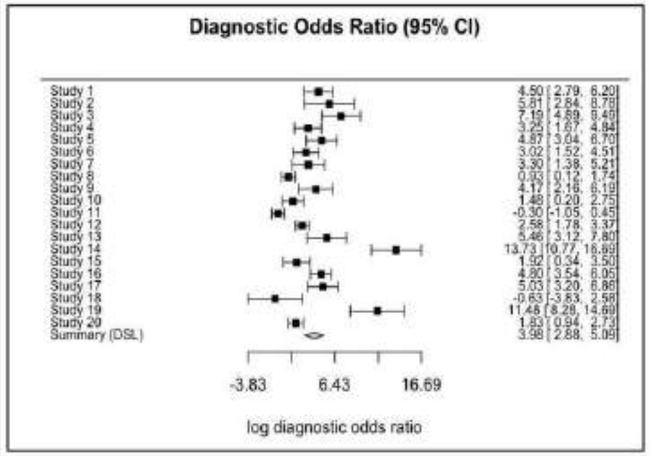
Application of Machine Learning with Convolutional Neural Network Method for Alzheimer’s Disease Detection: Systematic Review and Meta-Analysis





This project is a comprehensive systematic review and meta-analysis of the application of Convolutional Neural Networks (CNN) for early detection of Alzheimer’s disease using brain imaging, particularly MRI.

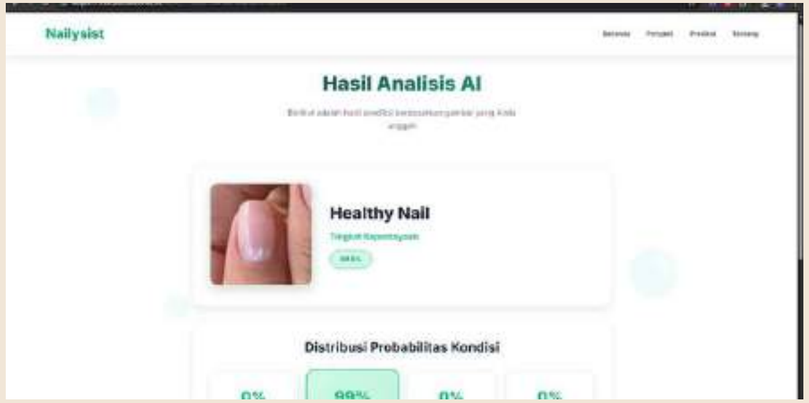
Early detection of Alzheimer’s is crucial for effective therapeutic intervention. The aim of this project is to evaluate the diagnostic performance of CNN models across various studies and to identify factors influencing their performance.

Skills: Meta-analysis · Machine Learning · Systematic Review

--> Link Projek <--

Mochammad Fahd Ali Hillay | 2026

NAILYSIST: Disease Detection from Nail Images Using Machine Learning



The "NAILYSIST" project is the development of an intelligent system based on deep learning and machine learning for early detection of systemic disease indications through nail image analysis. Nail conditions can serve as important indicators of overall health, but visual diagnosis is often subjective.

This system aims to provide a fast, accurate, non-invasive, and easily accessible early diagnostic aid, especially in areas with limited medical facilities.

Skills: Machine Learning · Deep Learning · Computer Vision · Image Processing

--> Link Projek <--

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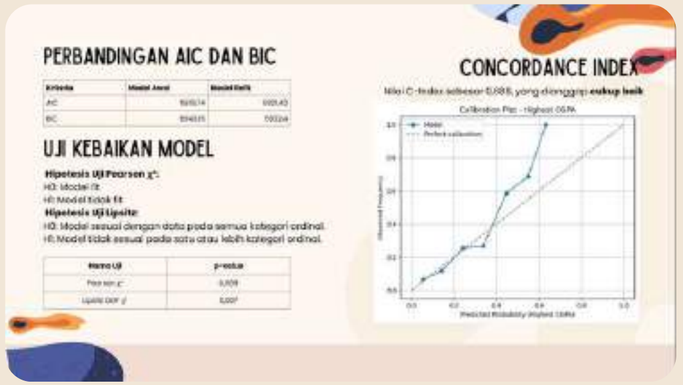
Ordinal Logistic Regression Analysis of Factors Affecting Student CGPA (Case Study in Bangladesh)



This project analyzes the factors influencing students' Cumulative Grade Point Average (CGPA) in Bangladesh, which serves as a key measure of academic achievement. Using an ordinal logistic regression model, the study examines demographic, academic, and psychosocial predictors of CGPA categories among 2,028 undergraduate and postgraduate students across 15 universities in Bangladesh.

Skills: Statistics · Exploratory Data Analysis · Python (Programming Language) · R (Programming Language) · Quantitative Analytics

--> Link Projek <--

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Quantitative Geospatial Analysis of Regional Socio-Economic Inequality Mapping in Indonesia

This project focuses on mapping and analyzing socio-economic inequality between provinces in Indonesia for the year 2024 using a quantitative approach. Our goal is to identify and understand socio-economic disparities, uncover key factors influencing poverty levels, and reveal spatial patterns to help prioritize development planning.

Skills: Geospatial Data · Quantitative Analytics · Exploratory Data Analysis · Data Analysis

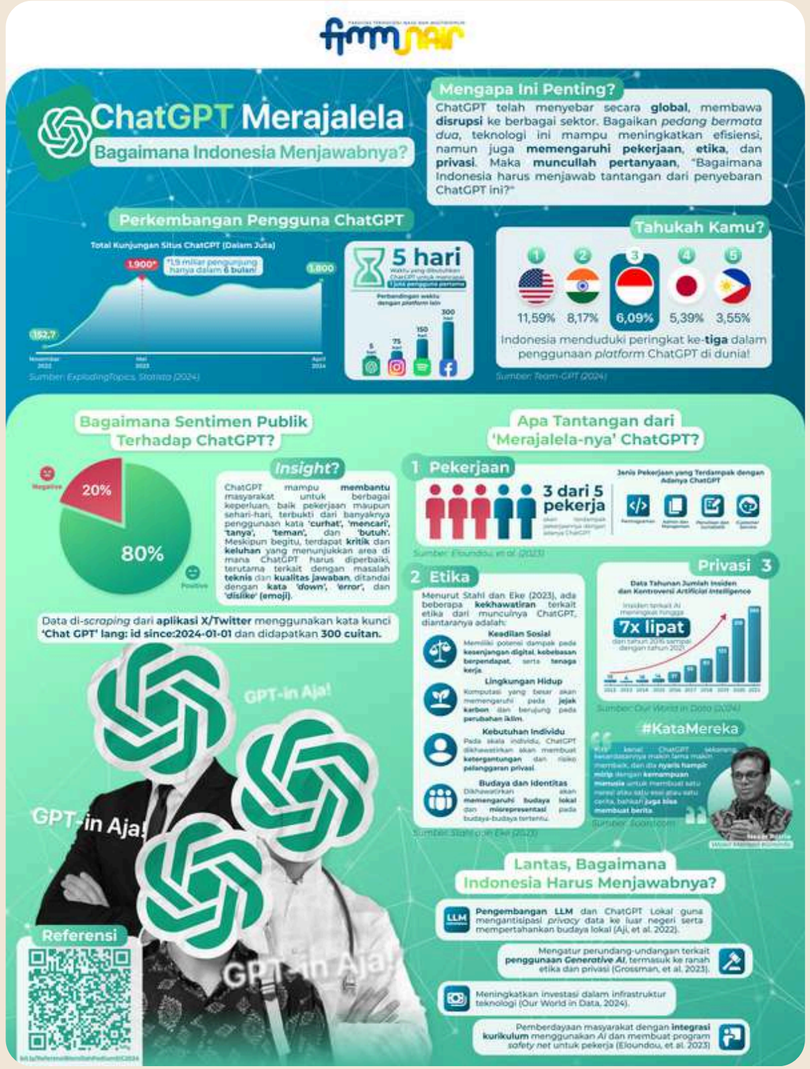
--> Link Projek <--

Mochammad Fahd Ali Hillay | 2026

DATA VISUALIZATION PROJECT'S

2023-2026

Mochammad Fahd Ali Hillay | 2025



PUBLIC SENTIMENT TOWARDS CHATGPT ON SOCIAL MEDIA

This sentiment analysis visualization illustrates how the public responds to ChatGPT on social media platforms like Twitter. Words such as 'helpful' and 'search' indicate positive benefits, while 'error' and 'down' highlight areas needing improvement. I created this visualization to analyze and present the public's sentiment, showcasing both the advantages and challenges associated with ChatGPT's adoption. This visual representation helps in understanding user experiences and identifying key areas for enhancement.

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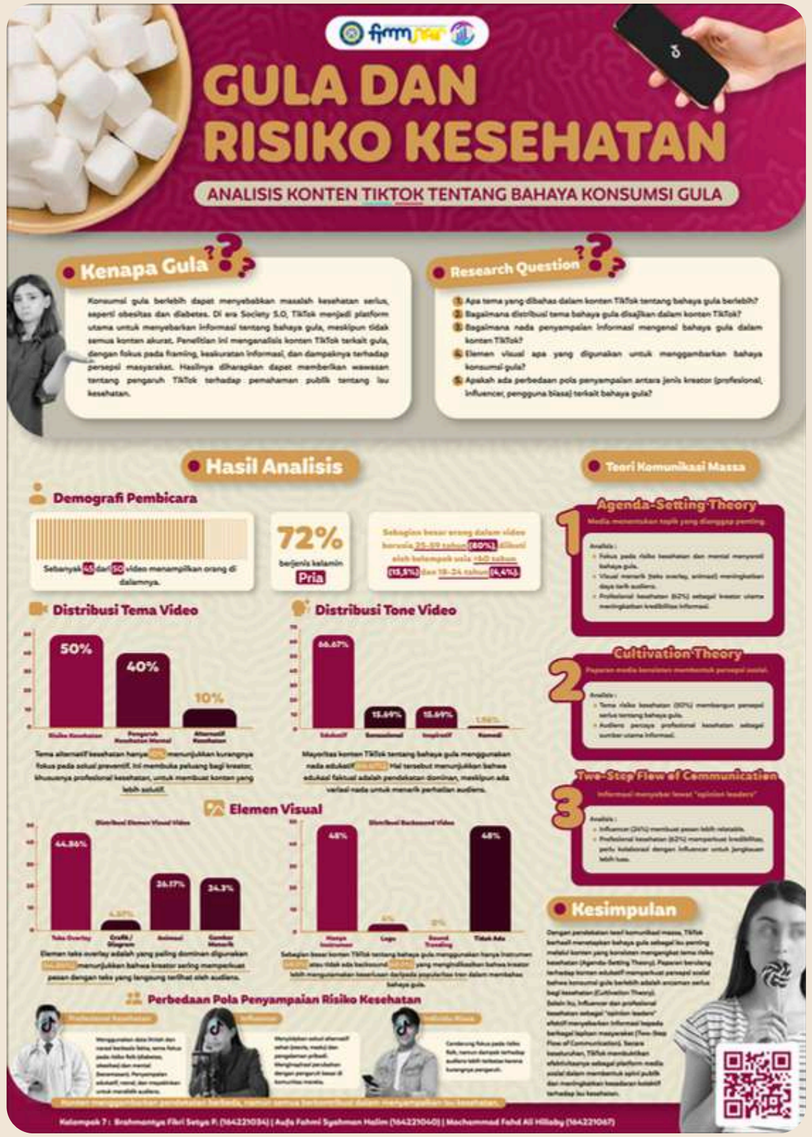
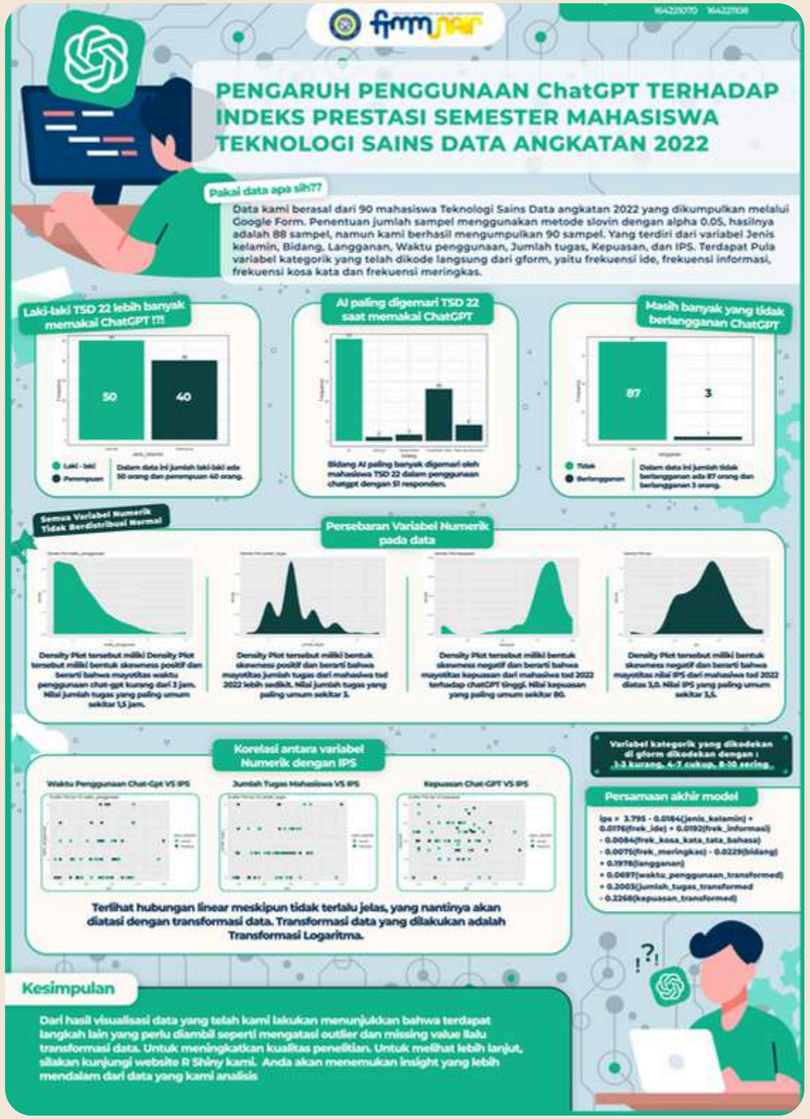


ANALYSIS OF TIKTOK CONTENT ON THE HEALTH RISKS OF SUGAR CONSUMPTION

This visualization analyzes TikTok content on the health risks of excessive sugar consumption. Key findings show a focus on women aged 25-59, with themes on health and behavior delivered in an educational tone. Videos use engaging visuals, with professionals emphasizing data and influencers promoting alternative solutions. Theories like Agenda-Setting and Cultivation highlight how TikTok shapes public perception, effectively establishing sugar risks as a critical issue and influencing societal behavior.

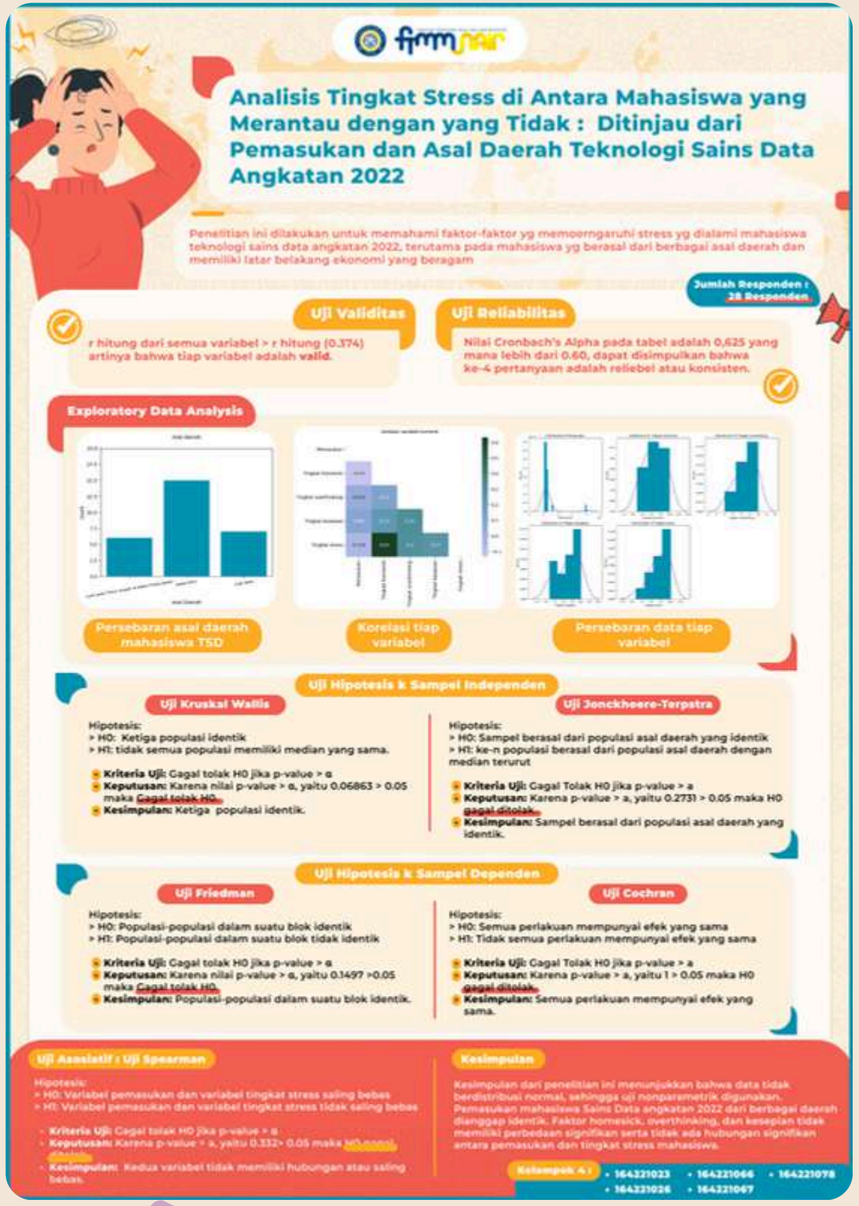
Mochammad Fahd Ali Hillay | 2026



ANALYSIS OF CHATGPT USAGE IMPACT ON ACADEMIC PERFORMANCE OF DATA SCIENCE STUDENTS

This visualization analyzes ChatGPT usage among 90 Data Science students (2022 cohort) and its impact on GPA. Data includes gender, field of interest, subscription status, usage time, task frequency, satisfaction, and GPA. Findings show most students use ChatGPT for under 3 hours daily, primarily for AI tasks, with high satisfaction (avg. 80) and GPAs above 3.0. Correlation analysis links task frequency, usage time, and satisfaction to GPA, with log transformations improving accuracy. The regression model identifies key factors influencing academic performance, highlighting ChatGPT's role in student success.

Mochammad Fahd Ali Hillay | 2026



ANALYSIS OF STRESS LEVELS AMONG MIGRANT AND NON-MIGRANT STUDENTS

This visualization explores stress levels among Data Science students (2022 cohort), comparing migrant and non-migrant students based on income and regional backgrounds. Data from 28 respondents was analyzed using non-parametric tests due to non-normal distribution. Key findings show no significant differences in stress levels across regions or income groups, with factors like homesickness, overthinking, and loneliness having minimal impact. Tests such as Kruskal-Wallis, Jonckheere-Terpstra, Friedman, and Spearman confirmed that stress levels and income are independent, and all treatments (stress factors) have similar effects. The study concludes that stress levels among students are consistent regardless of regional or economic differences.

Mochammad Fahd Ali Hillay | 2026

CERTIFICATE & ACHIEVEMENT

2023-2026

Mochammad Fahd Ali Hillay | 2026

CERTIFICATE

119587

PUSAT BAHASA DAN MULTIBUDAYA
UNIVERSITAS AIRLANGGA

03251200257

CERTIFICATE

This is to certify that

Mochammad Fahd Ali Hillaby

164221067

sat for **English Language Proficiency Test**

on

December 10, 2025

Best Score Obtained :

Subjects	Score
English Listening Section	49
English Structure Section	42
English Reading Section	53
Total Score	480

Valid To : **December 10, 2027**

Surabaya, **December 10, 2025**
Head,

Lina Puryanti, S.S., M.Hum., Ph.D.
NIP. 197301311998022002

PBMUA

COPIED COPIED COPIED

English Language Proficiency Test

CERTIFICATE

Data Science Course Level Basic

Data Science Course Level Intermediate

ACHIEVEMENT

WICE

Certificate of Award

is given to

Mochammad Fahd Ali Hillaby

from

Universitas Airlangga

IT and Robotics Category has been awarded the

SILVER MEDAL

for the project

"SI MENTARI": MENTAL HEALTH MANAGEMENT APPLICATION INTEGRATED ARTIFICIAL NEURAL NETWORK AND DATA SCIENCE TO REALIZE GOLDEN INDONESIA 2045

in World Invention Competition and Exhibition 2024
Online Competition

Selangor, Malaysia
September 21st – 25th, 2024

Dr.Nagaraja Suryadeavara — Head of Engaged Learning Experience and Innovative Teaching (ELEVATE) Department, MAHSA University, Malaysia

Deni Irawan, M.Pd — President of Indonesian Young Scientist Association (IYSA)

Razmi Bin Abdul Razak — President of Malaysia Young Scientists Organizations (MYSO)

Ganesan Jeyabalan — President of Malaysia Innovation, Invention & Creativity Association

Dipindai dengan CamScanner

Silver Medal World Invention Competition and Exhibition 2024

SERTIFIKAT

Nomor : 3592/PL19/KM.04.03/2024

Diberikan Kepada :

MOCHAMMAD FAHD ALI HILLABY

Sebagai

BEST PRESENTATION KATEGORI MAHASISWA

Business Plan Competition

Dalam Kegiatan Business and Maritime Festival 5.0
Himpunan Mahasiswa Manajemen Bisnis Politeknik Perkapalan Negeri Surabaya
Tingkat Nasional Yang Dilaksanakan Pada Tanggal 8 Juni 2024

Wakil Direktur III Politeknik Perkapalan Negeri Surabaya — Dr.Mohammad Haksam,S.T.,M.T. — NIP 197307032008011014

Ketua Himpunan Mahasiswa Manajemen Bisnis — Tami Abriansyah — NRP 1122040125

Ketua Pelaksana Business and Maritime Festival 5.0 — Driyanita Anggraini — NRP 1122040133

Best Presentation Business and Maritime Festival 5.0

ACHIEVEMENT

DIREKTORAT PEMBELAJARAN DAN KEMAHASISWAAN
DIREKTORAT JENDERAL PENDIDIKAN TINGGI, RISET, DAN TEKNOLOGI
KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI

Sertifikat

NOMOR 7308/E2/DT.01.00/2023

DIBERIKAN KEPADA

ANDINI MAULIDIYAH RACHMA
SOFIA NAISA HIDAYAT
MISFA SHAFWAH ZAHIDAH
AMELIA SAFIRA PUTRI VATURRACHMA
MOCHAMMAD FAHD ALI HILLABY

Universitas Airlangga
Sebagai peraih **pendanaan** bidang **PKMK**
NOMOA: Pemanfaatan Senyawa Eugenol dan Sitral pada Daun Salam Syzygium polyanthum sebagai Penangkal Kecoa

Program Kreativitas Mahasiswa (PKM) Tahun 2023

Jakarta,15 November 2023
Direktur Pembelajaran dan Kemahasiswaan

SRI SUNING KUSUMAWARDANI

NIP 196911221995122001

Peraih Pendanaan Program Kreativitas Mahasiswa (PKM) 2023

Direktorat Pembelajaran dan Kemahasiswaan
Ditjen Dikti Risti - Kemendikbudristek

SERTIFIKAT

Nomor: Manual.1144/E2/KM.01.00/2024

DIBERIKAN KEPADA

MISFA SHAFWAH ZAHIDAH
ANDINI MAULIDIYAH RACHMA
SOFIA NAISA HIDAYAT
AMELIA SAFIRA PUTRI VATURRACHMA
MOCHAMMAD FAHD ALI HILLABY
Universitas Airlangga
peraih **pendanaan** bidang **PKMK**
ODORE: Inovasi Produk Biodegradable Penghilang Bau dan Antibakteri Berbasis Pemanfaatan Limbah Tongkol Jagung (Zea mays) dan Ampas Kopi (Coffea sp)

Program Kreativitas Mahasiswa (PKM) Tahun 2024

Jakarta, 19 Agustus 2024

Sri Suning Kusumawardani

DIREKTUR PEMBELAJARAN DAN KEMAHASISWAAN

PKM AWARD 2024

Peraih Pendanaan Program Kreativitas Mahasiswa (PKM) 2024

Pusprestnas
BPTI

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
PUSAT PRESTASI NASIONAL
BALAI PENGEMBANGAN TALENTA INDONESIA

Sertifikat

Nomor 29249/BPTI/DIKTI/2024

Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi, Pusat Prestasi Nasional, Balai Pengembangan Talenta Indonesia menyampaikan penghargaan kepada:

MOCHAMMAD FAHD ALI HILLABY
Universitas Airlangga
atas prestasinya sebagai
Peserta bidang **PKM-Kewirausahaan**
pada **Pekan Ilmiah Mahasiswa Nasional (PIMNAS) ke 37 Tahun 2024** yang dilaksanakan oleh Balai Pengembangan Talenta Indonesia sebagai unit pelaksana teknis Pusat Prestasi Nasional, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi secara luring bekerja sama dengan Universitas Airlangga, Provinsi jawa Timur pada tanggal 14 s.d. 19 Oktober 2024.

18 Oktober 2024
Pit. Kepala Balai Pengembangan Talenta Indonesia,

Dr. Maria Veronika Irene Herdjiono, S.E., M.Si.
NIP 196103292012122001

PIMNAS 37

Peserta Pekan Ilmiah Mahasiswa Nasional (PIMNAS) 2024

GET IN TOUCH

Feel free to reach out!

+62 812 348 55470

instagram.com/fahdatauaby

linkedin.com/in/mochammad-fahd-ali-hillaby

Olivia Wilson | 2026

THANK YOU

Mochammad Fahd Ali Hillay | 2025